Set1

Experiment with Protege OWL

The pizza ontology is a well-known ontology in the semantic web community. It is developed for educational purposes by the University of Manchester, which is a leading university in the development of semantic technologies.

You can download it from here:
https://protege.stanford.edu/ontologies/pizza/pizza.owl

Open the pizza ontology in Protege.
Take some time to browse the class hierarchy, the property hierarchies and the individuals and note how the ontology describes the domain of pizzas.

Use DL Query

1. Get all pizzas from America

2. Get all pizzas with a spinach topping
   (SpinachTopping)

3. Get all pizzas with a FruitTopping
   (FruitTopping)

4. Get names of all pizza with MeatTopping
   (NamedPizza , hasTopping , MeatTopping)

5. How many direct subclasses does Pizza have?
   Can you modify the ontology so that IceCream is not?

Set 2

ex_tiny_4zebra.owl

1. Test ex_tiny.owl
2. Create an example with the operation complement (“NOT”) and show it works

Set 3: OWL solves a riddle

different than zebra puzzle

1. Choose a riddle from here:
   https://www.brainzilla.com/logic/zebra/einsteins-riddle/
   Try to solve it by hand. Present (/insert) the (partial) solution in <semigroup>in-class-activity.docx
2. Create an OWL solution.
   If it is too difficult, simplify the problem or try easier riddles (for example, see:
   https://www.brainzilla.com/logic/zebra/basic-1/)
   Create an OWL solution, then verify it works ! Upload the solution for Sem1: OWL solves a riddle assignment (The deadline is the date set on MSTeams assignment.)